

Quantization at the Entropy Bound: Calibration-Free Packing of At-Capacity Networks

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Abstract

Post-training quantization methods earn their complexity on some models and waste it on others, and current practice offers no way to tell which regime a model is in without running the quantizers. We give a zero-inference criterion. A network is *at capacity* when its weight code stream, on a per-row $\sigma/2$ grid, carries entropy equal to the Gaussian capacity of that grid; small math-task transformers trained to convergence sit within 1% of this bound at $n=3$ births on two architectures. At capacity there is little structure left for calibration to exploit: at matched 5 bits, GPTQ with a real Hessian, AWQ with real activation profiles, and HQQ all land within one gate sigma of a closed-form σ allocation. Off capacity, the premium those methods earn is monotone in a disk statistic $M = \text{span_bits} - \text{code_entropy}$ that costs nothing to compute: measured premiums run from $\sim 1\times$ at M 0.96–1.61 (our born crystals) through $16.3\times$ at M 2.85 (MoE experts) and $22\times$ at M 3.11 (attention) to $34\times$ at M 3.62 (one web-dense LLM). The meter has been read on five production model families (OLMoE, Qwen3-30B, DeepSeek-V3, Kimi-K2, Kimi-K3); only one web-dense premium is actually measured, and the web-dense band spans M 3.62–3.85. Around the criterion we ship: a closed-form packing artifact whose disk format is its runtime format (a bit-packed 5-bit Metal GEMV kernel, $2.39\times$ over fp16 at the best of five benchmarked shapes on synthetic Gaussian weights); a lossless rANS stage reaching $3.67\times$ over bf16 on a 30B MoE with zero calibration on a laptop; and a fixed-point decode path whose logit traces are bit-identical across GPU vendors and laboratories — including on a frontier model’s own shipped MXFP4 expert, consumed natively with zero requantization. Every experiment was pre-registered; exploratory reads are labelled as such, and the falsified predictions are reported alongside the confirmed ones.

1 Introduction

The quantization literature is a contest of increasingly sophisticated calibrated methods, each demonstrated on web-trained LLMs, each earning real margins there. This paper starts from an observation those demonstrations skip: on some networks, all of that machinery ties a two-line closed-form baseline to within the resolution of our gate.

The networks where calibration ties are not toys chosen to make the point. They are transformers trained to convergence on a verified task distribution — models we gate-score exactly, so parity claims are measured in solved problems, not proxy perplexity. On these models we measure (Sec. 2) that the weight entropy at a fixed quantization step equals the capacity of a Gaussian source at matched variance to within 1%. Nothing in the weights prefers any code over any other; there is little for a Hessian or an activation profile to find. We call such networks *at capacity*.

Web-trained LLMs are not at capacity, and the distance from capacity turns out to be measurable from the weight file alone. Our meter $M = \text{span_bits} - \text{code_entropy}$ prices exactly one thing: what a fixed-width uniform grid pays to reach the worst outlier in each row. Across every

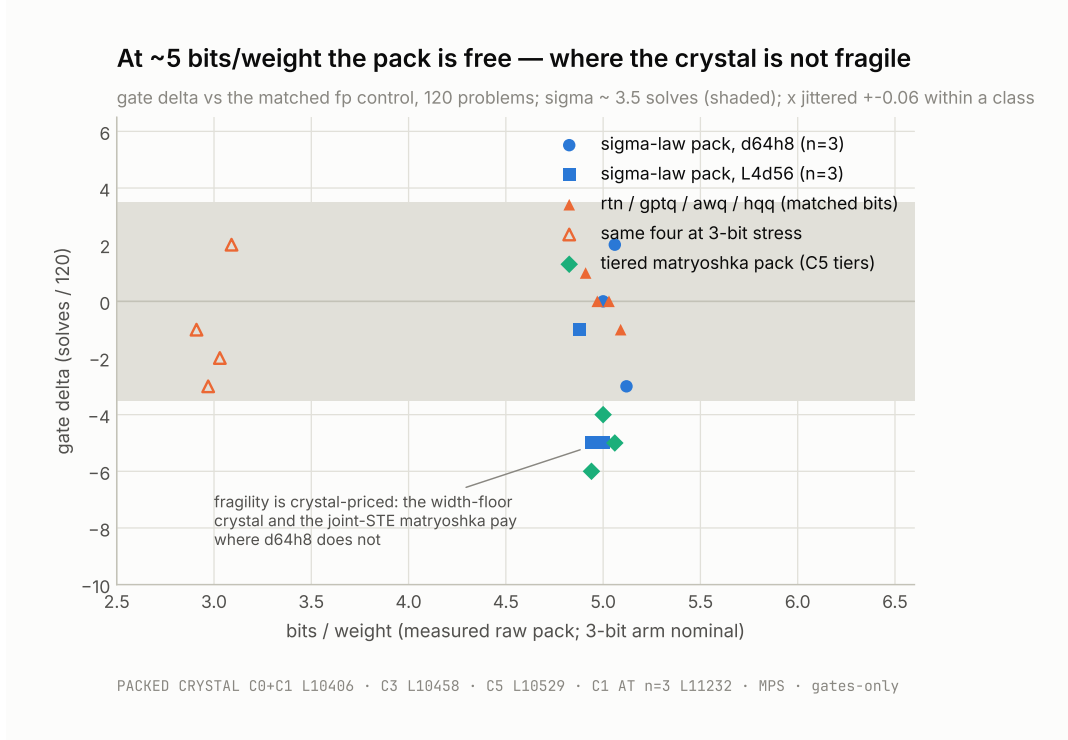


Figure 1: The packing curve across the C-series cells (C0/C1/C3/C5). The claim: on at-capacity crystals the closed-form σ pack sits at the entropy bound and calibrated methods do not separate from it at matched bits. Fence: MPS, $n=1$ per quantizer arm with C1 controls reused; the parity claim itself is $n=3$ and scoped to the d64h8 class.

group we metered — born crystals, MoE experts from five production families, attention blocks, dense web LLMs — the premium that calibrated methods earn over the closed form rises with M (Sec. 3). The practical consequence is a decision rule you can run before loading the model onto an accelerator: below $M \sim 2$, use the closed form and keep the calibration budget; above it, per-row max-anchored grids and their calibrated refinements earn their cost.

Three artifacts fall out of taking the entropy bound seriously (Secs. 4–6): a packed format at ~5 bits/weight whose gate score holds within sigma on our headline class at $n=3$; an entropy-coded container within a rounding error of the measured code entropy, lossless, at 30B scale on a laptop; and — because integer codes admit exact arithmetic — a decode path that produces bit-identical logit traces on Apple and NVIDIA silicon, replicated cross-laboratory, and demonstrated on a frontier MoE’s own shipped 4-bit expert format.

Sections 7–9 add three measurements that constrain what the entropy account can and cannot see: bits are blind to the *geometry* of the grid, the *symmetry* axis carries a separate and priceable toll, and the meter behaves as a predictive instrument with named misses. Section 10 is the honest-limits section — where the method fails, by how much, and at what n .

We report our falsified predictions with the same prominence as the confirmed ones. Several of them sharpened the criterion into its final form; one closes an entire class of post-hoc MoE compression and redirects the effort to the systems layer (Sec. 11).

2 The entropy bound, measured

Fix a weight tensor W with per-row standard deviation σ_r . Quantize row r on a uniform grid of step $\sigma_r/2$ and read the resulting integer codes as a symbol stream. Two numbers describe that stream: its empirical entropy H , and the capacity of a Gaussian source quantized on the same grid, $H^* = \frac{1}{2} \log_2(2\pi e) - \log_2(\text{step}/\sigma)$. If the weights were drawn i.i.d. Gaussian, H would equal H^* ; any learned structure — correlations, heavy tails, clustering — shows up as $H < H^*$ (compressible structure) or as grid overhead (outliers stretching the span).

The choice of $\sigma/2$ is set by a separately measured precision knee, not by the entropy argument. Across three crystals and two widths the knee sits where the grid step is $0.5\text{--}1.0\sigma$: at $d=56$ ($\sigma \approx 0.13$), $Q=16$ is free (grid 0.48σ) and $Q=8$ bites (0.96σ); at $d=256$ ($\sigma \approx 0.06$), $Q=64$ is free (0.25σ) and $Q=16$ bites (1.0σ). The knee constant was later hardened to $n=3$ paired births: $Q=64$ within ± 3 of comparator $3/3$ ($0/-1/0$) and $Q=16$ at -7 or worse $3/3$ ($-12/-13/-9$), at near-identical sigmas $0.0367\text{--}0.0369$. So $\sigma/2$ is a step comfortably inside the free zone, not “the coarsest step that keeps parity.”

Measured on our born crystals (d64 8-head and layers-4 d56 math-task transformers), H sits within 1% of H^* : code entropy 3.10–3.13 against capacity ≈ 3.13 on all four birth/architecture cells at $n=3$. This is the paper’s anchor measurement: these networks are *at capacity* — maximum-entropy on their own grid, with little left for a smarter code to find.

The at-capacity state has a sharp operational consequence, verified directly in Sec. 4: every calibrated quantizer we tested lands inside the gate’s resolution of the closed form. And it has a converse: any model whose code stream falls short of capacity, or whose span outruns its entropy, is *not* at capacity, and the shortfall is exploitable — by exactly the methods the literature already built. The rest of the paper measures that gap and the toolchain that lives on its floor.

3 The meter: regime detection from disk

$M = \text{span_bits} - \text{code_entropy}$, computed at per-row $\sigma/2$, no forward passes. span_bits is what the fixed-width grid must pay per weight; code_entropy is what the weights actually carry. Their difference prices the worst-case outlier against the typical weight — mechanically the same thing a calibrated outlier-aware method gets paid for handling well.

Table 1 is the instrument. Two caveats travel with it and we state them here rather than in a footnote. First, the “NNUE 0.82” row is an in-passing assertion made in the C7 verdict text, not a cell with its own receipt; we keep it only with that label. Second, exactly *one* web-dense premium is measured (Qwen2.5-0.5B at $M\ 3.62 \rightarrow 34\times$); SmolLM2-1.7B at $M\ 3.85$ was metered but never given a σ -vs-HQQ premium arm, so the web-dense band is $3.62\text{--}3.85$ in M with a single premium anchor.

Two falsifications built this instrument, and we report them as such. First, “strong transport” — the claim that σ allocation would hold on any expert-shaped tensor — failed on OLMoE experts ($16.3\times$ premium to HQQ at matched 6 bits); the meter had flagged the miss before the arms ran, in an amendment written at kill time, which is what promoted it from diagnostic to dial. Second, kurtosis, the obvious moment statistic, is demoted by measurement: the OLMoE expert group read kurtosis 3.50 — nominally Gaussian — while carrying $M\ 2.85$ worth of grid overhead.

The dial comes with a second condition, also learned from a falsification: zero-tax deployment requires an evaluation metric with knee slack. Under outcome scoring (does the model still solve the task), our packs are free below $M \sim 2$; under per-token KL, nothing is free — on OLMoE experts the dial policy read $\Delta\text{KL}\ 0.0187/\text{tok}$ against RTN’s 0.0097 , i.e. $1.9\times$ worse. We argue

Table 1: The dial. M at per-row $\sigma/2$, param-weighted, alongside the measured calibration premium (σ -allocation ΔKL divided by HQQ’s ΔKL , same device, matched bits). Premiums are measured only where a premium column is given; the rest are meter-only rows.

group	M	kurtosis	measured premium
crystal cplx_none	0.96	2.26	—
crystal d64h8	1.61	3.70	$\sim 1\times$ (ties at gate)
crystal L4d56	1.61	3.56	—
house NNUE	0.82	—	—
Kimi-K3 experts ($\sim 33\text{M}$)	1.94–2.15	3.12	—
Kimi-K2 experts ($\sim 40\text{M}$)	2.01	3.02–3.04	—
DeepSeek-V3 experts ($\sim 45\text{M}$)	2.33	3.07	— (predicted 5–10 $\times$, unmeasured)
OLMoE experts ($\sim 6\text{M}$)	2.85	3.50	16.3 $\times$
Qwen3-30B experts ($\sim 5\text{M}$)	2.93 (median)	—	—
OLMoE attention	3.11	7.55	22 $\times$
Qwen2.5-0.5B dense	3.62	5.29	34 $\times$
SmolLM2-1.7B dense	3.85	6.54	—

outcome scoring is the deployment-realistic metric, and we say explicitly that this is an argument, not a theorem.

Expert size, and where the trend breaks. A scaling regularity was proposed on the MoE rows: expert capacity tracks *per-expert size*, not expert count — Qwen3-30B ($\sim 5\text{M}/\text{expert}$) at M 2.93, OLMoE ($\sim 6\text{M}$) at 2.85, DeepSeek-V3 ($\sim 45\text{M}$) at 2.33, Kimi-K2 ($\sim 40\text{M}$) at 2.01. We state the tension in the same breath: this ordering is *not* monotone in size. The 45M-parameter DeepSeek experts read 2.33 while the smaller 40M Kimi-K2 experts read 2.01, and Kimi-K3’s $\sim 33\text{M}$ experts read 1.94–2.15, straddling K2’s 2.01. The K3 row additionally carries a grid-image confound: K3 ships MXFP4, so the meter reads the shipped 4-bit image, which caps span and biases M low. The honest statement is a *band* — large experts sit near $M \approx 2.0$, at the σ -law boundary — not a rank ordering.

Frontier practice is consistent with the band. Kimi-K3 ships 896 experts *per layer* across 93 layers in a latent MoE (experts read a 3584-dim latent, so an expert is $\sim 33\text{M}$ parameters, not the $\sim 66\text{M}$ naive estimate — a figure we established by pulling a single expert from the 2.8T release by HTTP byte-range). Its shipped 4-bit codes carry 3.643 bits/param, so MXFP4 leaves roughly 9% lossless margin: Moonshot packs within ~ 0.36 bits of its own format’s capacity.

The atlas cell shows the meter at production scale: Qwen3-30B, 30.2B parameters across 18,673 tensors, metered and dial-packed in a 136.6-minute laptop pass with zero calibration. Routers are the incompressible organ (M 4.45, all top-5 function errors are routers — keep them fp); expert up_proj reads median M 1.93, already inside the σ -law domain, against gate/down at 2.93. The mid-stack dip the atlas found (layers ~ 8 –16 at M 1.93) did *not* generalize: Kimi-K2 reads flat in depth (2.09/2.04/2.09), so the dip is a recipe or scale artifact, not a law.

4 The artifact: packing at the bound

The packing rule is two lines. For each 2-D block tensor t with standard deviation σ_t , choose the integer grid $q_t = \lceil 2/\sigma_t \rceil$ — i.e. step $\sigma_t/2$ — round, and store the codes bit-packed at the width the

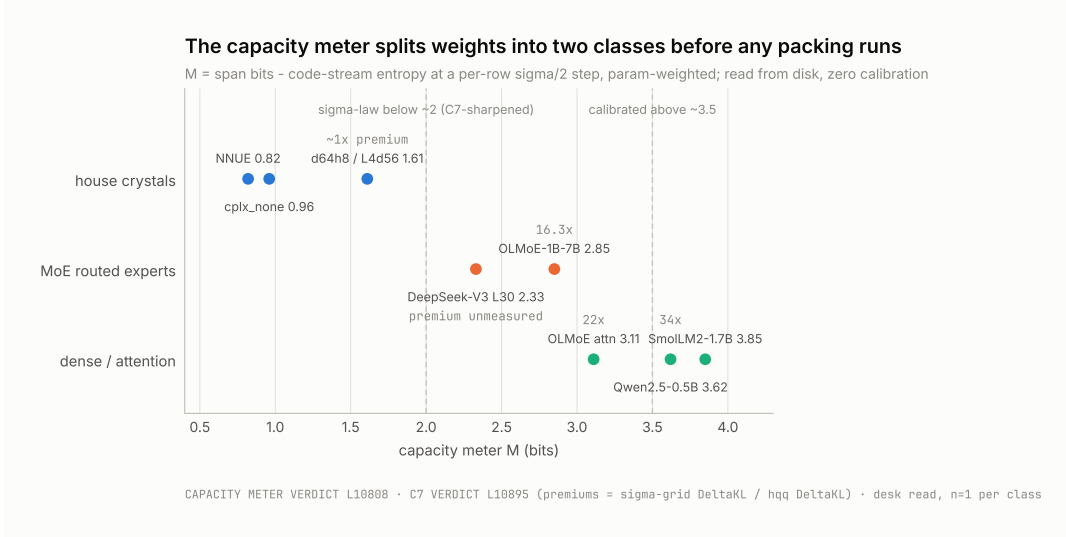


Figure 2: The capacity meter as a dial. The claim: the calibration premium rises with M across the measured groups. Fences: premiums are single-model, $n=1$ per arm, same-device paired; the DeepSeek row’s 5–10 $\times$ is a *predicted*, unmeasured premium and quantized-release rows (K2, K3) read the shipped grid’s image, not the fp master.

span requires (~ 5 bits/weight on our models). Embedding, head and norm tensors stay fp32 in the booked artifact: they are tiny and were never snapped. (A separate $\sigma/8$ interface-tensor knob was tried in the fixed-point decode work and changed nothing; we do not ship it.) No calibration data, no optimization, no inference.

Parity. On the d64h8 class, the pack holds gate parity within sigma at $n=3$: seed 1 $58 \rightarrow 58$, seed 2 $54 \rightarrow 56$ (+2), seed 3 $59 \rightarrow 56$ (−3) — all three deltas below the gate sigma of 3.5. Seed 1 is exact; the class-level claim is parity within sigma, not exactness. On the layers-4 d56 depth-floor class the pack pays −5 at both new seeds ($50 \rightarrow 45$, $52 \rightarrow 47$) against seed 1’s −1; two readings beyond sigma is not noise. We therefore scope the zero-tax claim to the d64h8 class and name per-crystal fragility (Sec. 10) as the axis the entropy argument does not see. Compression is 6.15–6.65 $\times$ over fp32.

The tie. At matched 5 bits on at-capacity weights, GPTQ with its real Hessian (58), AWQ with real activation profiles (58), HQQ with its per-tensor optimization (57) and plain RTN (59) all land within sigma of both the fp control (58) and our pack (58); ΔKL runs 0.0011–0.0018/tok. This is an underpowered null at $n=1$ per arm, not a demonstration of equality: the correct reading is that no calibrated method separates from the closed form at this resolution. A 3-bit stress arm shows where damage lands when it does come: solve rates stay flat (rtn 57, gptq 55, awq 56, hqq 60 — the pre-registered $>\sigma$ drop did *not* fire) while validity falls from 57% to 46–54% and ΔKL jumps 20–40 $\times$.

Wall-time is calibration time. The pack runs in 0.9s where HQQ takes 61.7s at 0.5B scale (69 $\times$), and 16.6s where HQQ takes 675.5s on 6.4B expert parameters (41 $\times$). Two honest qualifications. First, these are *calibration-time* numbers, not inference-time. Second, and more important: both cells come from C6 and C7, which are precisely the regimes where our transport claim *failed*.

Where parity actually holds — the 200k–500k-parameter crystals — calibration is sub-second (0.9 s pass, <0.2 s optimize) and the wall-time differentiator is trivial. The speed gap is real but it is not evidence for the closed form in the regime the closed form wins.

The disk format is the runtime format. A Metal GEMV kernel consumes the bit-packed 5-bit codes directly — six signed codes per uint32, no unpack pass. Across five benchmarked shapes it reads $1.11\times (224\times 54)$, $1.21\times (256\times 66)$, $1.07\times (1536\times 384)$, $1.73\times (8192\times 2046)$ and $2.39\times (14336\times 4098)$ over fp16, beating the byte-aligned variant of the same kernel at the same shape class ($1.76\times$): the denser format wins on bandwidth. The $2.39\times$ headline is the best of those five shapes, on synthetic Gaussian weights at crystal sigma (0.19), $n=1$ per shape. The honest losses: at micro shapes the kernel is parity-within-noise — a 256×64 cell read $0.91\times$ once and $1.03\times$ on replication, and micro-shape entries are fenced accordingly. The one clean “first attempts lost” story belongs to a different kernel line: the int4 dequant-GEMV ladder went $0.47\text{--}0.70\times$ (v1), $0.75\text{--}1.00\times$ (v2), $1.11\times$ at $D=4096$ (v3).

One negative fences the artifact class. On a jointly-trained matryoshka (tiered) crystal, every tier of the nested pack pays 1–2 sigma. Tiered packing is real (–15% bytes per solve under escalation economics) but its zero-tax version requires packing EMA parents, not joint-STE children. Fragility, again, is orthogonal to entropy ($n=1$).

5 Lossless coding: the bound as bytes

If the code stream carries H bits/weight, an entropy coder should store it in H bits/weight, and the meter already measured H from disk. A range coder (rANS) over the σ -law codes achieves that. The pre-registered prediction was that streams land within 0.5% of entropy, and it confirmed — but the *artifacts* carry table overhead on top: the d64h8 artifact is 233,180 bytes ($9.10\times$ over fp32) at 3.179 bits/wt against a code entropy of 3.12, i.e. +1.9%; L4d56 reads $8.25\times$; all six seed crystals sit in a 3.16–3.21 bits/wt band. Tables add roughly 2% on the small crystals, counted honestly; there is no 0.1% claim.

At scale: the packed-plus-coded Qwen3-30B artifact is 16.48 GB (codes 4.337 bits/wt + 0.09 GB scales) against the 60.4 GB bf16 release — $3.67\times$ smaller, lossless, produced on a laptop in one pass with zero calibration. Losslessness is asserted per stream, with the first 2B Qwen symbols explicitly roundtrip-verified and coder identity relied on thereafter. The pipeline composes: rANS is the at-rest container, and decompression yields the same bit-packed codes the Sec. 4 kernel executes, so the storage format and the execution format are two states of one object. Fences: $n=1$ per artifact; decode-side rANS throughput is not benched.

6 Determinism: exact arithmetic as a deployment property

Integer codes admit exact arithmetic, and exact arithmetic is order-invariant: an int64 sum gives the same bits under any reduction schedule, which is precisely what floating-point inference cannot promise across GPUs. We build this out in three stages, each verified by hash equality, not tolerance.

Stage 1 — exact GEMMs. Weights as integer codes, GEMMs on an exact-fp32 carrier with hi/lo splitting keeping every partial below 2^{24} (the fp32 mantissa bound, asserted at runtime). The cell books *two hash channels*, not two seeds: hash A (integer GEMM over all σ -law code tensors) is bit-identical on MPS and cuda, while hash B (the fp32 full-forward logits) differs across the same

pair; greedy streams match anyway, per the near-tie doctrine. A replication pass re-ran hash A on a second *activation* battery seed and it reproduced identically.

Stage 2 — full fixed-point decode. RMSNorm as int64 sums-of-squares with integer-Newton isqrt; SiLU, softmax-exp and RoPE as integer tables generated once and shipped as bytes (libm variation across platforms is exactly the thing being excluded, so tables must travel, never be regenerated). The complete per-step logit *traces* — every number, not just the argmax — hash identically on MPS and cuda, with max GEMM partial $2^{21.2}$ against the 2^{24} bound. The capability price, measured honestly: 96.66% teacher-forced argmax agreement against the fp parent over 3,055 tokens, with disagreements concentrated at coin-flip margins (median 0.177 vs 7.6 overall); the reference implementation runs 10–40 $\times$ slower than fp, and speed is not the claim. An independent laboratory reproduced both digests from the sha-pinned tables file on first attempt, giving one tables file across cpu, MPS (two machines) and cuda, in two labs, with no tolerance column. This cross-lab leg covers the Stage-2 (P3) result only, and the second lab’s contribution was its MPS leg.

Stage 3 — a frontier model’s own format. Kimi-K3’s routed experts ship as MXFP4: e2m1 integer codes times power-of-two scales — already exact. We pulled one expert (17.5 MB by safetensors byte-range, out of a 2.8T release), consumed the shipped codes natively with zero requantization, and measured sha-identical integer-GEMV traces on cpu, MPS and cuda for all three matrices. A follow-up cell closed the composition fence: the full expert *forward* (w1/w3 GEMVs $\rightarrow$ power-of-two requant $\rightarrow$ shipped SiLU table $\rightarrow$ gate $\times$ up $\rightarrow$ w2 GEMV) hashes identically on the same three backends. Both K3 cells are house-only, on a synthetic 64-vector battery, and are an *expert forward*, not a full-model decode. Two experts were metered; one was hash-locked at both GEMV and chain level.

7 Bits are geometry-blind

The entropy account makes a strong implicit claim: what a quantization grid costs a network depends on the *information* the grid destroys, not on the coordinate system the grid is drawn in. We tested that directly, four ways, and it survived all four.

Polar versus Cartesian grids. On a complex-FFN crystal (cplx_none, $d=384$, control 63), we swept uniform σ -step grids against polar grids that quantize magnitude and angle separately. The uniform arm obeys the σ -priced knee as predicted (0.5σ free at 63, 2σ bites to 60). The pre-registered prediction that polar wins at matched measured bits was falsified: 4.75-bit polar reads 61 against 4.62-bit uniform’s 60 — parity, sub-sigma, not a win. The angle-starvation prediction was falsified too: *eight* angle bins are free at fine magnitude (63 solves at 6.23 bits/weight). The law that books: the snap knee follows total measured bits/weight regardless of grid geometry.

Rotating the grid under star-shaped weights. The obvious objection is that cplx_none’s weights are isotropic, so no coordinate frame is special. We re-ran on a “star” crystal (cplx_G5_dep, control 66) whose weights are strongly anisotropic. A 45-degree grid rotation moves every nonzero star weight by $|\delta c| \approx 0.77|c|$ — massive distortion — and cost 1 solve against the aligned grid: polar4 aligned 63, polar4 rotated-45 62, uniform 63. The aligned-free/misaligned-craters prediction was falsified. Anisotropic weights do not imply axis-priced quantization.

Table 2: The symmetry ladder. Each rung projects a trained crystal’s gate matrices onto a group’s commutant, then heals one epoch. Comparator 65/120, $n=1$ per arm, gates-only, MPS.

rung	group	sharing	projected init	healed gate	anti-mass
R3	complex	$2\times$	57	64	2×10^{-4}
S1	quaternion	$4\times$	22	61	7×10^{-4}
S4	Z_2	$2\times$	49	64	2×10^{-4}
S3	C_8 (block-circulant)	$8\times$	2	59	2.8×10^{-3}

No spontaneous rotational structure to align to. A complementary read explains why. Measuring anti-commutant mass against an adjacent-pairing complex structure, every real crystal sits inside the random-pairing null band at ≈ 0.500 (wfloor 0.49983, s2 0.50022, muon 0.50045, 19m 0.50021) — there is no hidden rotational subspace for a rotated grid to respect. Projecting the gate matrices onto the commutant gives a strikingly gentle damage curve: $t=0$: 65, $t=0.25$: 65, $t=0.5$: 64, $t=1.0$: 57 — deleting 50% of gate mass into the commutant costs 8 solves. One warm epoch under a ramped commutation penalty heals that back to 64 at anti-mass 2×10^{-4} .

The alphabet does not have to follow the domain. The sharpest version of the geometry question was run on a ZX-calculus diagram task, where the data itself carries phases: does a rotational (complex) weight alphabet beat a scalar magnitude alphabet on phase-carrying data? Rotational read 32/120 against magnitude’s 31/120 with a pre-registered bar of ≥ 5 ; alphabet-follows-domain is dead. The same tie holds on the math diet ($62 = 62$).

Fences for this section: single-seed cells, MPS (the ZX column excepted, cuda), and gate/up matrices only.

8 The symmetry axis and its toll curve

Bits are one compression currency. *Structure* — imposing a group symmetry on the weights so that k parameters share one degree of freedom — is another, and it prices differently. We measured a ladder of impositions on the same trained crystal (comparator 65/120), each by the same recipe: project the gate matrices onto the commutant of the group, then heal for one warm epoch under a ramped commutation penalty.

The toll curve reads $2\times$: -1 solve, $4\times$: -4 , $8\times$: -6 (Table 2). The remarkable cell is S3: dense gate matrices of a trained crystal retrofit into block-circulant (convolution) structure at $8\times$ parameter sharing, healing from a projected init of 2/120 to 59/120. The no-spontaneous-structure null holds at all three groups tested (complex, quaternion, Z_2/C_8) — these symmetries must be imposed, never found.

Exactness is available, at a width price. Doubling the crystal’s width ($d\,256 \rightarrow 512$, $W \oplus W$ everywhere, head $[W/2|W/2]$, heads $4 \rightarrow 8$ at fixed head_dim) produces an *exactly* complex-linear model: all eight doubled gates assert anti-mass $< 10^{-12}$, and the gate reads exactly 65/120 — the comparator, with no floating-point tie flipped. So both endpoints of the trade are measured: exact at $2\times$ width (cost: $4\times$ params) versus healed at $1\times$ width (cost: ~ 1 solve, params/2). Compression and exactness pull opposite ways on this axis.

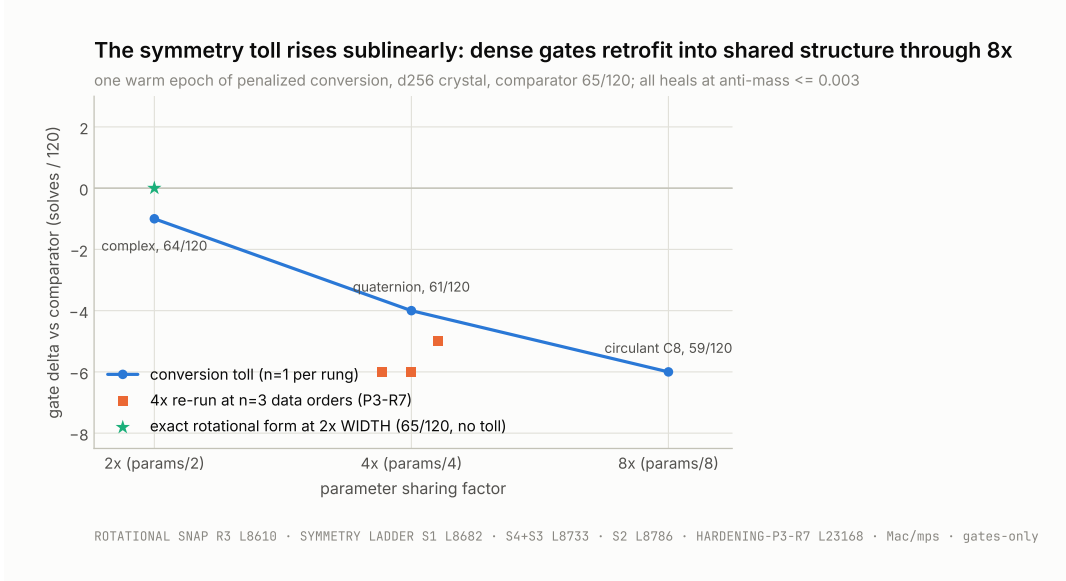


Figure 3: The symmetry toll curve (R3, S1, S3, S4, S2, and the P3-R7 hardening). The claim: imposing a $k\times$ parameter-sharing group symmetry on a trained crystal costs a small, roughly increasing number of solves, and an exact form exists at $2\times$ width for the complex case. Fence: gates-only, MPS, $n=1$ per rung except the $4\times$ rung, which is $n=3$ data-order seeds (pooled -17).

The $4\times$ toll, hardened. The quaternionic rung was the one worth $n=3$. Against comparator 65, three data-order seeds read 59 (-6), 59 (-6) and 60 (-5), pooled -17 , 3/3 negative, every heal at anti-mass $\leq 7 \times 10^{-4}$. The $4\times$ toll is therefore best quoted as **5–6 solves** at $n=3$, not the single-seed -4 . (The replication variable had to be found: the torch seed measured inert in this driver, so the hardening seeds redirect the driver’s hardcoded data-order shuffle.)

The axes are not orthogonal — except when they are. Two cells bracket the interaction. On the 19M crystal, quantization and sharing compete for one slack budget: at $Q=16$ the dense substrate pays -12 (65 to 53) while the circulant- $8\times$ substrate pays -25 (59 to 34), a delta-of-deltas of -13 against a gate sigma of 3; at $Q=8$ both are destroyed. But a later cell on the star lineage found the opposite for a different pairing of axes: $4\times$ quaternionic sharing changed the ternary-sensitivity profile by nothing — retention fractions of each arm’s *own* base are equal class by class (emb 62% vs 58%, head 102% vs 100%, attn 0% vs 0%, ffn 26% vs 26%). The reconcilable reading is that the compression corner measures a shared slack *pool* at a fixed harsh grid, while the ternary profile measures the *shape* of sensitivity across locations; sharing consumes the former and leaves the latter untouched.

A third asymmetry is worth naming because it rules out a tempting story: Z_2 and complex both share parameters $2\times$, but their projected inits differ sharply (49 vs 57) — equal mass removed, unequal damage. Structure, not mass, is what the toll prices.

Discussion: the division-algebra reading is an analogy, not a measurement. It is tempting to read Table 2 as a Hurwitz ladder — $\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$ at $1\times, 2\times, 4\times, 8\times$ — with the toll as the price of losing commutativity and then associativity. We label that framing explicitly as analogy, and we hold our own counterexamples against it. The $8\times$ rung is *not* octonions: it is C_8 block-circulant structure, a cyclic-group convolution, which has nothing to do with the octonion multiplication

table. And Z_2 and complex sit at the same $2\times$ sharing with different damage, which a division-algebra account does not predict. The measured object is a toll curve in parameter sharing; the algebraic story is decoration on top of it.

9 The capacity meter as a predictive instrument

The meter was pre-registered as a predicate before its read: at step $\sigma_r/2$ per row, $M = \text{span bits minus code-stream Shannon entropy, param-weighted over 2-D weights, with three named predictions and a shipped decision rule either way}$. We record how it scored, because a meter’s value is its calibration, not its existence.

- **Prediction 1 (crystals in 1.5–2.5): confirmed** — crystals read 0.96–1.61, in band and if anything cleaner than predicted.
- **Prediction 2 (web-dense ≥ 4): directional, not hit** — Qwen2.5-0.5B 3.62 and SmolLM2-1.7B 3.85. The separation from the crystal band is clean, but the numeric threshold was missed by about 0.3 and we book that as a miss.
- **Prediction 3 (MoE experts closer to crystals than to web-dense): confirmed** — DeepSeek-V3’s routed experts read kurtosis 3.07 and M 2.33.

The instrument’s real test came at C7, where it was used *forward*. Packing OLMoE’s expert tensors at matched 6 bits, $\sigma[\text{row}]$ read ΔKL 0.0716/tok against rtn 0.0097 and hqq 0.0044 — a $16.3\times$ premium to HQQ; the dense attention control read 0.0442 vs rtn 0.0053 vs hqq 0.0020, a $22\times$ premium. The strong-form transport prediction missed: OLMoE experts are not crystal-band (M 2.85, kurtosis 3.50 — Gaussian-shaped but wide-spanned). Crucially, the meter said so *before* the arms ran: the kill-time amendment recorded experts at M 2.85, 0.35 above the predicted band. That is the sense in which the premium is monotone in M — and the sense in which kurtosis is demoted: 3.50 looked Gaussian, and span-priced M caught what the fourth moment missed. The decision rule sharpened accordingly: σ -law only below $M \sim 2$. The wall-time prediction confirmed at scale: HQQ 675.5s on 6.4B expert parameters against σ ’s 16.6s, $41\times$.

The whole C-series was replicated on every arm: the C0/C1 rerun reproduced pack stats and all four gates line-for-line; C4’s cross-device integer invariance held on a fresh activation battery; C6c’s ΔKL values reproduced to printed precision; large-shape kernel speedups landed within 2% of booked.

Exploratory: an open lead at $M \approx 1.96$. A V4-Flash read puts a two-expert sample at $M \approx 1.96$. This is *exploration grade*: $n=2$ experts, no RESULTS booking, artifact rows in `data/lake/weights.parquet` (L1E107, L29E13). We report it as an open lead and draw no conclusion from it.

10 Measured limits

This section is not an appendix. The method has a boundary and we measured where it is.

The falsifier fired on the first real LLM. Per-tensor σ allocation does not transport to Qwen2.5-0.5B. Against an fp16 control perplexity of 60.29, the σ pack (avg raw 6.82 bits, matched to 7) read ΔKL 1.2839/tok and ppl 96.39, against rtn7’s 0.1669/60.90 and hqq7’s 0.0388/60.52. That is $33\times$ HQQ’s ΔKL — the pre-registered falsifier bar was $5\times$, and it fired decisively. Even plain RTN, with per-row max scales, beats per-tensor σ by $8\times$.

The granularity rescue failed. The obvious rescue — keep the closed form but move to per-row grids — did not work: σ -pack[row] read ΔKL 1.3072/tok and ppl 90.26, statistically unchanged from per-tensor. Prediction 1 falsified: no $\geq 10\times$ recovery. Both arms anchor the step at $\sigma/2$, the crystal knee, and web-trained Qwen’s fragility constant is far larger than any crystal’s.

The step rescue works, and still loses. Moving to step $\sigma/8$ ($q_r = \lceil 16/\sigma_r \rceil$, 9.64 raw bits matched to 10) recovers $11.6\times$: ΔKL 0.1127/tok, ppl 60.64. The σ *form* survives; the knee constant, not the allocation form, was the failure. But rtn10 reads 0.0029 and hqq10 0.0007 — max-anchored grids own heavy tails, and prediction 2 (within $2\times$ of rtn) was falsified. The mechanism is now clean: our fixed-width bits are priced by the worst outlier’s span, so one 50-sigma outlier inflates every row’s bit count, while max-anchored grids spend the same bits as a finer step on every quiet row. Among *zero-calibration* methods, RTN is the one to beat, and on web-dense weights we do not beat it; HQQ (0.0044) in turn beats RTN (0.0097) on the OLMoE expert arm.

Attacking the span is catastrophic. If outliers inflate the span, clip them. We pre-registered a σ -clip allocator on SmolLM2-1.7B at 6 bits and it was destroyed: against an rtn(absmax) baseline of ΔKL 3.32 / ppl **62.98**, $k=4$ read ΔKL 8.82 / ppl **138,890**; $k=6$ gave ppl 5,913; $k=8$ gave 532; hqq read 0.445 / 61.96. Saturating even the $> 8\sigma$ tail kills the model. Web-dense outliers are load-bearing at full magnitude, and the span attack is dead.

There is no universal distortion curve. Recomputing 21 historical snap cells onto a normalized distortion axis D/σ^2 , the cells are monotone *within* each crystal but there is no universal $f(x)$: at $x \approx 0.09$ the 19M crystal keeps 0.878 of its solves while cplx_none keeps 1.000; d56 bites by $x \approx 0.014$ where cplx is untouched at $x \approx 0.12$ — a $\sim 30\times$ fragility spread at matched normalized distortion. The equation is two-parameter: kept $\approx f(k_c \cdot D/\sigma^2)$, with f a universal monotone shape and k_c a per-crystal fragility (rough fit: cplx_none 1, 19M ~ 4 –5, d56 ~ 25 –30). Geometry-blindness survives this (polar and Cartesian cells interleave on the same crystal’s curve), but the knee *constant* is crystal-priced — and k_c is exactly what the entropy meter cannot see.

The zero-tax claim’s scope. Zero tax means: the d64h8 crystal class, at $n=3$, under *outcome* scoring, below $M \sim 2$. It does not extend to the L4 depth floor (-5 at two seeds), to per-token KL as the metric (rtn wins), or to web-dense weights (Sec. 10, first three paragraphs).

Fence inventory: what is $n=1$ and what is not. $n=3$: the C1 pack-parity headline on d64h8 and the entropy-bound gap; the precision knee constant; the quaternionic $4\times$ toll; the init-default causal leg for expert decorrelation. $n=2$: the causal micro-MoE experiment (two seeds, house scale, one task diet); the cross-device integer hash (two activation batteries). $n=1$: every quantizer arm in C3; every meter row; every kernel shape; the C5 tier tax; the C6/C6b/C6c cells; the P2a-v2 clip arms; each symmetry-ladder rung except the $4\times$; and both K3 cells.

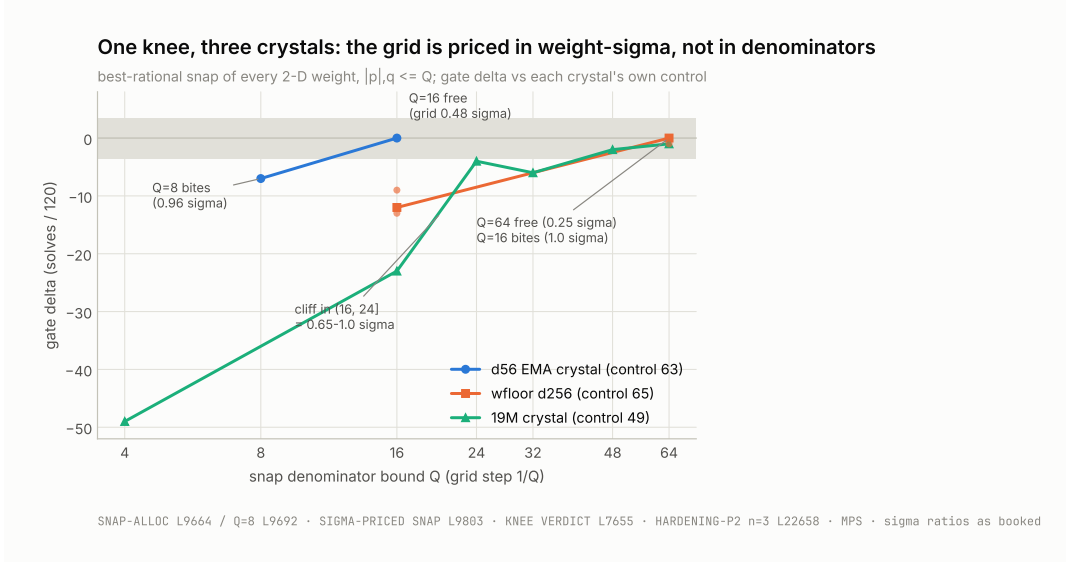


Figure 4: The precision knee in σ units. The claim: the free/bites boundary sits at a grid step of roughly $0.5\text{--}1.0\sigma$ regardless of width — d56 $Q=16$ free (63/63/63 vs baseline 63) and $Q=8$ biting ($-4/-4/-7$); d256 $Q=64$ exactly free (65/65/65); 19M sweep control 49 with $Q=16$ 26, $Q=24$ 45, $Q=32$ 43, $Q=48$ 47, $Q=64$ 48. Fence: the original cells are $n=1$, MPS, gates-only; the knee constant itself is hardened to $n=3$ paired births ($Q=64$: $0/-1/0$; $Q=16$: $-12/-13/-9$).

11 What routing does to weights

The MoE rows of the dial ($M \sim 1.9\text{--}2.9$) invite an obvious post-hoc program: find the redundancy between experts and compress it — merge similar experts, share bases, factor deltas. We measured the preconditions and closed the program.

In weight space, production experts are decorrelated to numerical zero. On Qwen3-30B, 128 `gate_proj` experts decoded at layers 12 and 40 read $\sigma(\text{delta})/\sigma(W) = 0.995/0.993$ with mean pairwise correlations of 0.0024 and 0.0054 (max 0.0155) — there is no shared component for a base-plus-delta factorization to extract. This was a *confirmed* house prediction, not a falsification: the pre-reg predicted ratio > 0.8 and correlations ≈ 0 , and both landed.

But the redundancy did not vanish; it moved. Co-routing mutual information between expert pairs *within* a layer runs 0.21–0.46 bits against a token-shuffle scale of 0.0009 — roughly $300\text{--}500\times$ baseline, at every layer. Merging the top-MI pair (layer 15, experts 1 and 29 — the most co-routed pair, not the most similar in weight space) costs perplexity $75.74 \rightarrow 79.10$, i.e. $+3.4$. Storage-side redundancy became usage-side structure — a systems lever (prefetch, placement, caching, as the offloading literature already exploits), not a parameters lever.

The causal experiment, and what it actually taught. We trained matched micro-MoEs (0.9M parameters, four experts, top-1 switch routing, two seeds) under (a) the standard load-balance loss, (b) no balance loss, and (c) tied base-plus-delta experts. Removing the balance loss changed neither the decorrelation (0.0080 vs 0.0085) nor the routing structure ($256\times$ vs $288\times$ shuffle), and the router did not collapse (worst share 55%). Both readings replicated at seed 2 (0.0082 vs 0.0085; MI $205\times$ vs $299\times$).

Our first mechanism for this was that *sparse assignment itself* creates the split — hard-routed experts never see the same tokens and so cannot stay correlated. That mechanism is retracted. A

soft-mixture arm, in which every expert sees every token, landed correlation 0.0089, indistinguishable from the top-1 arms; the prediction of a $\geq 10\times$ rise under soft routing was falsified, which kills the sparsity explanation. The desk check that settles it: expert correlation *at initialization* is 0.0016. Independently parameterized high-dimensional weight vectors start near-orthogonal and training raises them only to ≈ 0.008 in every regime tested; no routing regime we ran supplies a force toward correlation. Decorrelation is the **init default preserved**, not a product of routing. This causal leg is now $n=3$ soft-routing seeds (0.0096, 0.0071, 0.0062, all far below the $10\times$ -over-0.0016 falsification criterion), on one device.

Two riders. The production signature reproduces at 1/30,000th production scale (corr ≈ 0.005 , MI 205–299 $\times$ shuffle on a 0.9M-parameter micro-MoE), making the phenomenon cheap to study. And experts *tied at birth* were booked as a **split** prediction, not a clean win: the tied arm gates 43 / 42, the best MoE arm and within sigma of the dense control at both seeds, so the tie pays almost nothing — but the base does *not* dominate (deltas grew to 0.76–0.8 $\times$ base norm) and the deltas stay $\sim 7\times$ more correlated than untied experts. The factorization that post-hoc analysis proves impossible is approximately free if imposed before training, and it does not compress the way the tie was supposed to.

A seed-1 exploratory rider on (M , MI) opposite motion was killed honestly: seed 2 showed no monotone relation and the MI ordering across arms flipped (205 $\times$ lb / 299 $\times$ free at seed 2, against 288 $\times$ / 256 $\times$ at seed 1).

12 Related work

Calibrated PTQ. Our relationship to GPTQ [1], AWQ [2] and HQQ [3] is not competitive but *conditional*: on web-dense LLMs they earn a 16–34 $\times$ premium over our closed form and we say use them; on at-capacity networks they do not separate from it at gate resolution, and the meter tells you which case you are in from the file alone, with no forward pass. We are not aware of a prior pre-quantization regime test, though we make no claim to have surveyed the space exhaustively.

Outlier methods. LLM.int8() [4] and SqueezeLLM [5] rest on the premise that outliers are load-bearing. Our clipping falsifier is direct evidence for that premise from the opposite direction (clipping web-dense weights at 4σ is catastrophic — perplexity 138,890 against an rtn baseline of 62.98), and M is precisely a measure of the fixed-width price those outliers impose.

Information-theoretic codebooks. NF4’s quantile codebooks [6], QuIP#’s Hadamard incoherence processing [7], AQLM’s learned lattices [8] and TurboQuant’s random rotations [9] are a family of transforms *toward* the Gaussian regime. Our measurement is that convergent training on verified tasks produces that regime natively, so we characterize the endpoint of the line the transforms are walking.

Weight entropy coding. Deep Compression [10] Huffman-codes pruned CNNs and DeepCABAC [11] arithmetic-codes them. Section 5 is the LLM-scale revival, with the coding gain predicted in advance by the same statistic that selects the quantization regime.

MoE structure. Switch [12] and DeepSeekMoE [13] argue that fine-grained experts specialize; Section 11 is the weight-space receipt. Expert offloading systems [14] implicitly bet on routing locality; our co-routing MI is the statistic they are betting on, measured. REAP [16] appears here as bibliographic context only: we did not run it, and our merge probe is a single $n=1$ cell, so the

most we can say is that our result is not in tension with their report that merging carries irreducible error versus pruning.

Determinism. Integer-only inference exists for edge CNNs [15]. We are not aware of a prior LLM decode with cross-vendor bit-identical logit traces, nor one executing a frontier model’s shipped format exactly; we state this as the limit of our awareness rather than as a survey result.

13 Fences

Every scope limit, in one place.

1. The zero-tax parity claim is scoped to the d64h8 class at $n=3$ ($+2/-3/0$, all sub-sigma against gate sigma 3.5; exact only at seed 1); the L4 depth floor pays -5 at both weak seeds. Per-crystal fragility (k_c , near-tie flip density) is an orthogonal axis the entropy meter does not see.
2. The meter on quantized releases (Kimi-K2, Kimi-K3) reads the shipped grid’s image, not the fp master; the DeepSeek-V3 row is likewise a *dequantized* read (fp8 block-dequant, one shard). Those rows support bands, not rank orderings.
3. All paired comparisons are same-device, same-seed. The $2\times$ device-dependence figure belongs to the *L9 probe* (the fp32 champion scored 18/24 on cuda vs 9/24 on MPS at the near-tie frontier); at fp32 the production *gate* measured a device delta of -1 (MPS 70 vs cuda 69, sub-noise). We do not compare gates across devices regardless.
4. The determinism claim is about the fixed-point path’s own reproducibility; it is not fp equivalence (96.66% argmax agreement, disagreements at coin-flip margins), and the reference implementation is $10\text{--}40\times$ slower than fp.
5. The cross-lab replication covers the Stage-2 (P3) fixed-point decode only. The K3/MXFP4 cells are house-only across three backends; the second lab’s leg was MPS.
6. The Kimi-K3 cells are two experts metered and one hash-locked at GEMV and full-chain level on a synthetic battery — an expert *forward*, not a full-model decode.
7. Wall-time wins in Sec. 4 are calibration-time, not inference-time, and both cells come from regimes where our quality claim failed; where parity holds, calibration is sub-second.
8. Kernel speedups are $n=1$ per shape on synthetic Gaussian weights; the headline is the best of five shapes and micro shapes are parity-within-noise.
9. Sections 7 and 8 are single-seed, gates-only, MPS (ZX column: cuda), except the $4\times$ toll at $n=3$.
10. Cells at $n=1$ are named as such where they appear; see the fence inventory in Sec. 10.

14 Reproducibility

Every experiment in this paper was pre-registered in an append-only log — predictions and falsifiers named before the run fired. Reads taken outside a pre-registration are labelled exploratory in

the log and in this text (the (M, MI) rider and the V4-Flash lead are the two that appear here). Every falsification appears in the text with the same prominence as the confirmations; corrections are amendments naming their target, never edits. All instruments, seeds, and the log itself are in the repository. Numeric artifacts travel sha-pinned, and the replication protocol of Sec. 6 (pinned artifact, pinned instrument, full-digest compare) was validated by an independent laboratory reproducing both hashes on first attempt.

Where a result depends on a generated dataset, the generator uses stable string seeds and exclusion-guarded train/eval splits. One contamination incident is booked in the log itself: Holdout v1 was voided by its own audit at 281 corpus collisions on the 88M band. Two further incidents that motivated the same discipline (a mathgen L1/L2 eval-in-train overlap, and a generator whose `pick()` admitted only four distinct bodies) are recorded in the repository’s working-conventions file rather than in the results log; we name the location so the count is not overstated.

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